



Alternating Optimization for LLM Post-Training: From Divergence to Verified Solutions

The Chinese University of Hong Kong — Department of Computer Science and Engineering

jianghuanyun

Supervisor: Prof. Hoi To Wai

ALS & its 3 repairs → DIVERGE | SGD / AdamW / LoRA → CONVERGE | proof below (matched budgets, pretrained baselines shown)

Research Question

Why does ALS-based post-training diverge on deep LLMs, and what works instead? ALS solves $W^* = (X^T X + \lambda I)^{-1} X^T Y$ in closed form — one solve replacing hundreds of gradient steps. It converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt. Two sub-questions: (Q1) can the divergence be explained and fixed? (Q2) what actually works, compared fairly?

The Story in 4 Steps

STEP 1 — ALS diverges on deep models

The closed-form solve $W^* = (X^T X + \lambda I)^{-1} X^T Y$ converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt (11/11 on Qwen2.5-7B; PPL → NaN in 1–3 cycles).

STEP 2 — Mechanism proven by controlled ablation

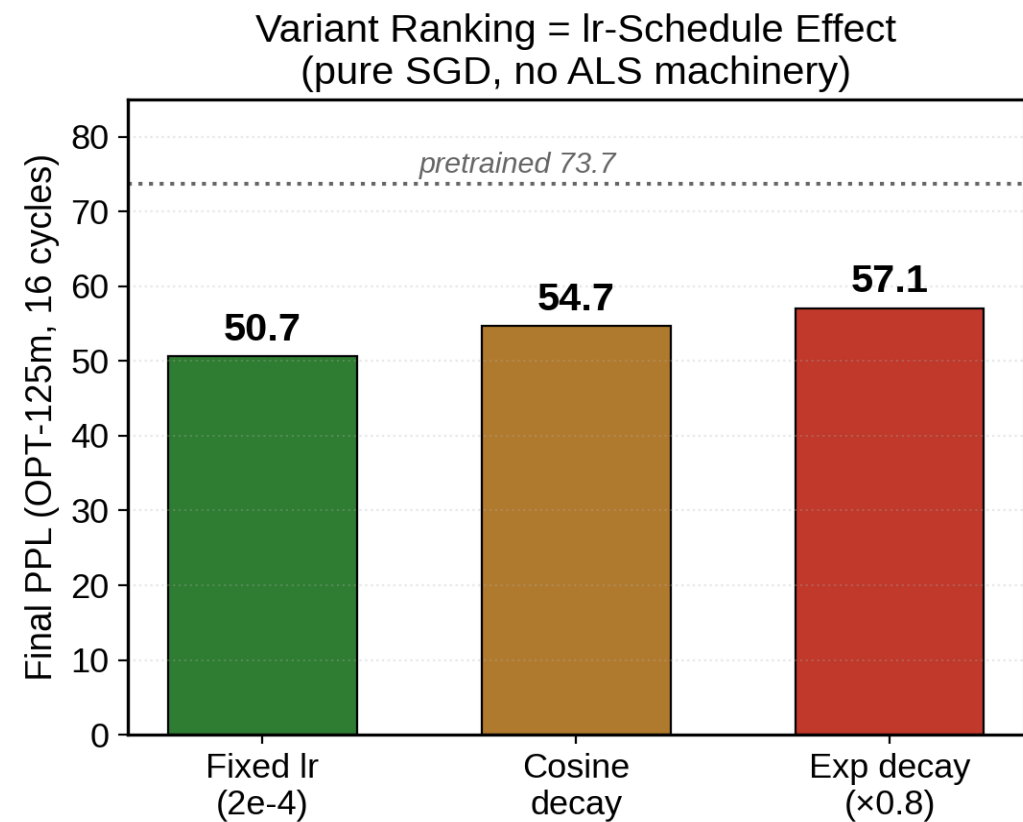
Five-condition ablation on Qwen2.5-7B isolates **ALS weight modification** as the necessary-and-sufficient cause within the tested space: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge (Fig. 5).

STEP 3 — All three natural repairs fail matched controls

Gradient injection, low-rank probing and soft-target distillation — each matched against an equal-budget SGD baseline — add no measurable value (Figs. 1–3).

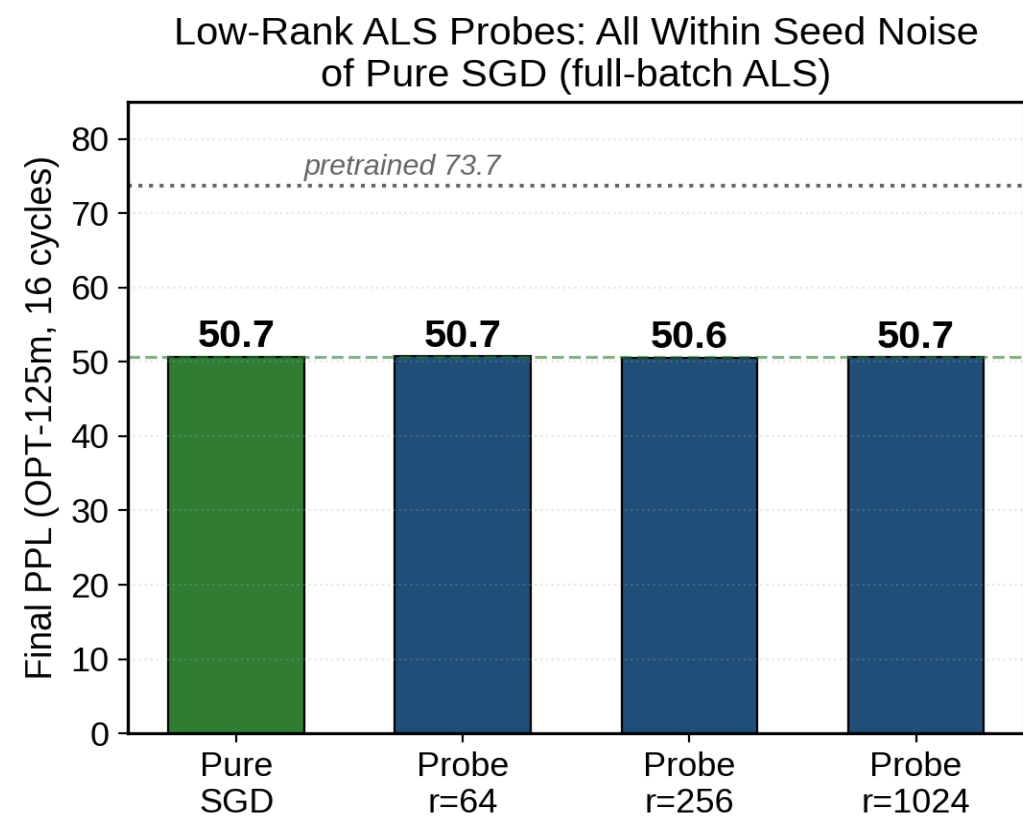
Finding 2 — Variant ranking = learning-rate-schedule effect

With injection vacuous, the ranking is explained by the lr schedule alone (pure SGD, no ALS machinery): fixed $50.7 < \text{cosine } 54.7 < \text{exp } 57.1$ — same order as the 7B variants.



Finding 3 — Low-rank ALS probes: within seed noise at every rank

$r = 64 / 256 / 1024$ with full-batch ALS solves: 50.7 / 50.6 / 50.7 vs pure SGD 50.7 — no rank improves on SGD. The closed-form solve is redundant with the CE-gradient direction.



STEP 4 — Verified alternatives, baselines shown

AdamW full-rank 1.25 PPL; AdamW+LoRA 10.41 PPL at 0.1% trainable params; plain SGD 6.83 PPL. All far below the pretrained 73.1 PPL reference — **up to 58× improvement over doing nothing** (Fig. 4).

Contributions

- **Diagnosis** — first controlled evidence that ALS weight modification (not noise / optimizer / hook overhead) causes deep-model divergence, via a 5-condition ablation.
- **Theory** — residual-amplification model ($\rho \approx 1.08/\text{layer}$) predicts a 25–28-layer boundary, consistent with 8 architectures.
- **Saved design space** — three plausible ALS repairs proven ineffective by matched-budget controls; identifies why (closed-form solve $\equiv$ CE gradient).
- **Verified alternatives** — AdamW 58×, SGD 10.7×, LoRA 7× below the 73.1 pretrained baseline, with a corrected evaluation harness.

Why ALS diverges — causal theory

Residual connections amplify an ALS perturbation by an **effective $\rho \approx 1.08$ per layer** (phenomenological fit), i.e. $\approx 8 \times$ over 27 layers — beyond SGD's per-cycle recovery. The predicted boundary falls **between 25 and 28 layers**, consistent with the observed 24-converges / 28-diverges split.

The perturbation *grows* across cycles ($\delta: 0.085 \rightarrow 0.196$) — a positive feedback loop, which is why damping/clipping patches only delay the failure.

Method & discipline

- **2×2 factorial** (ASP vs AdamW × full-rank vs LoRA), FLOPs-matched, shared seeds/data.
- **Harness audit**: OPT numbers re-verified with correct tokenizer + pad-masked labels (baseline 73.7 PPL); 7B baseline 73.1 PPL.
- **Falsification discipline**: pre-stated hypothesis, open code, negative results reported with matched controls.

Verified Solutions (Qwen2.5-7B, 28 layers)

AdamW full-rank — 1.25 PPL (N=3, 800 steps). Best absolute quality.

AdamW + LoRA — 10.41 PPL at $\sim 0.1\%$ trainable params. Lowest trainable-parameter cost.

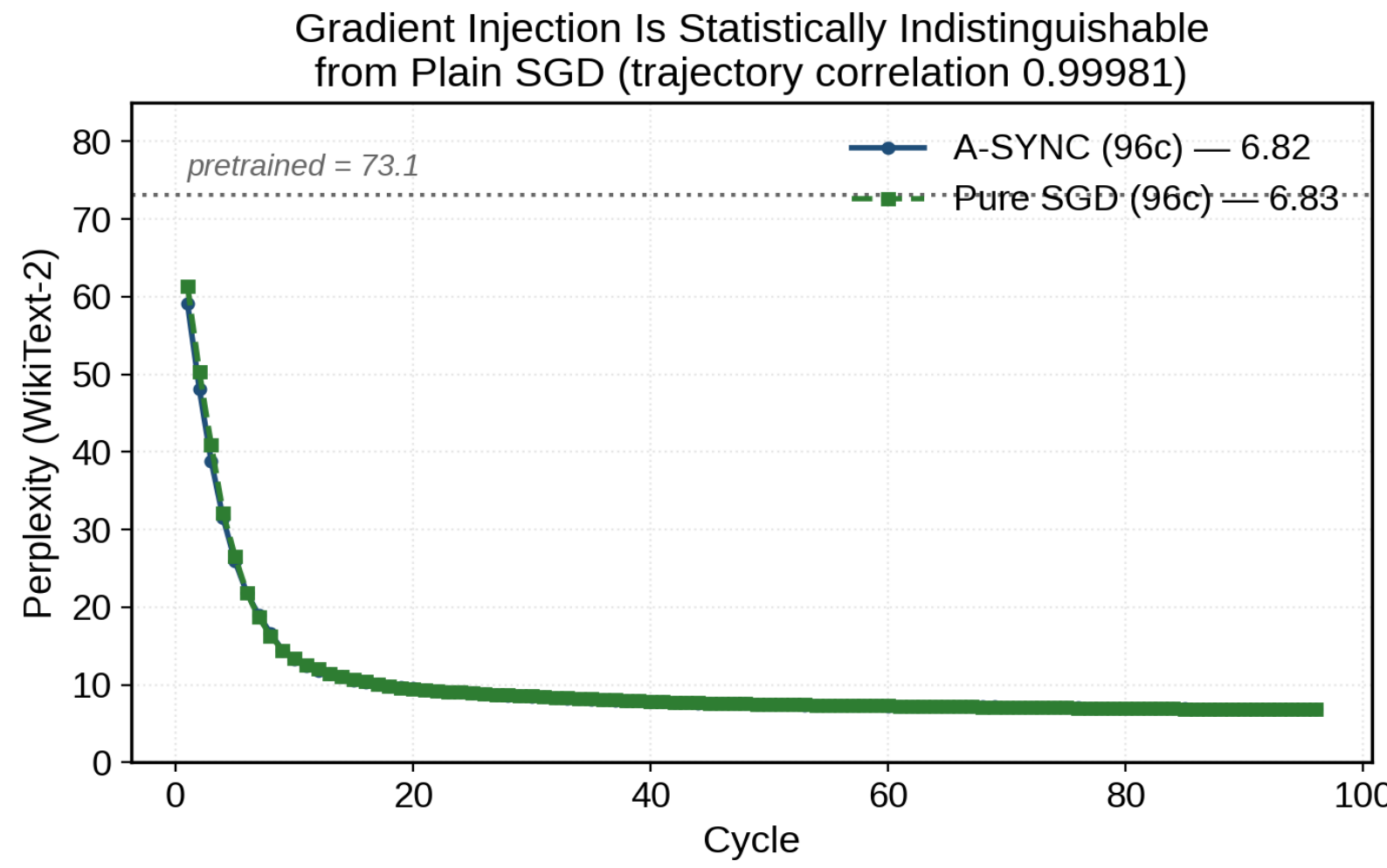
Plain SGD, fixed lr — 6.83 PPL in 4800 steps (power-law tail). The ALS-free baseline that converges where ALS cannot.

Practical takeaway: on deep models, remove the ALS machinery and let gradient descent do its job with a sustained learning rate — 58× better than the pretrained baseline with AdamW, 10.7× with plain SGD. Best quality still requires AdamW.

Key Findings & Experimental Results

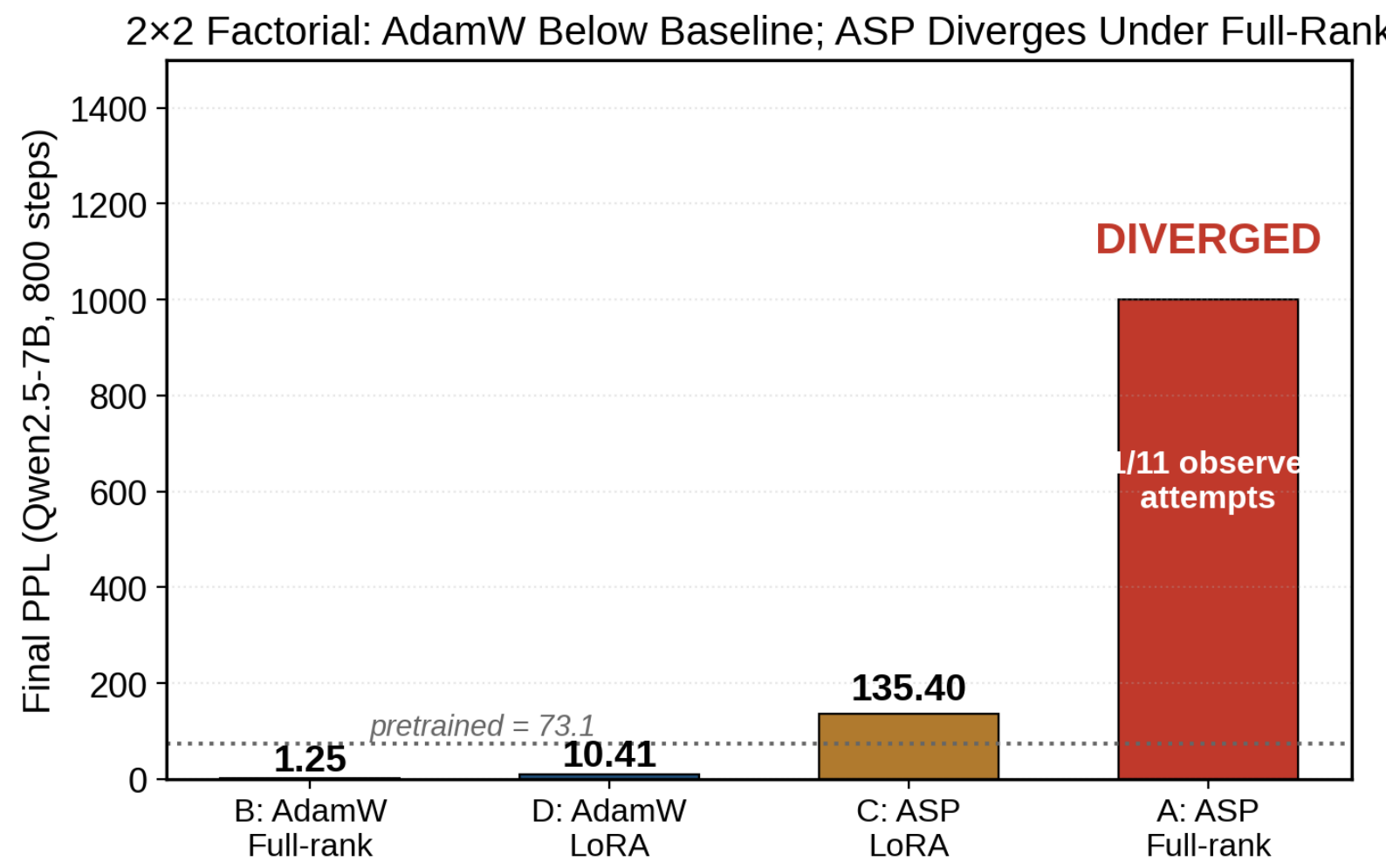
Finding 1 — Gradient Injection: statistically indistinguishable from plain SGD

A-SYNC (96c) 6.82 vs pure SGD (96c) 6.83; trajectory correlation 0.99981, mean per-cycle difference 0.116 PPL. *No tested schedule beats plain SGD.*



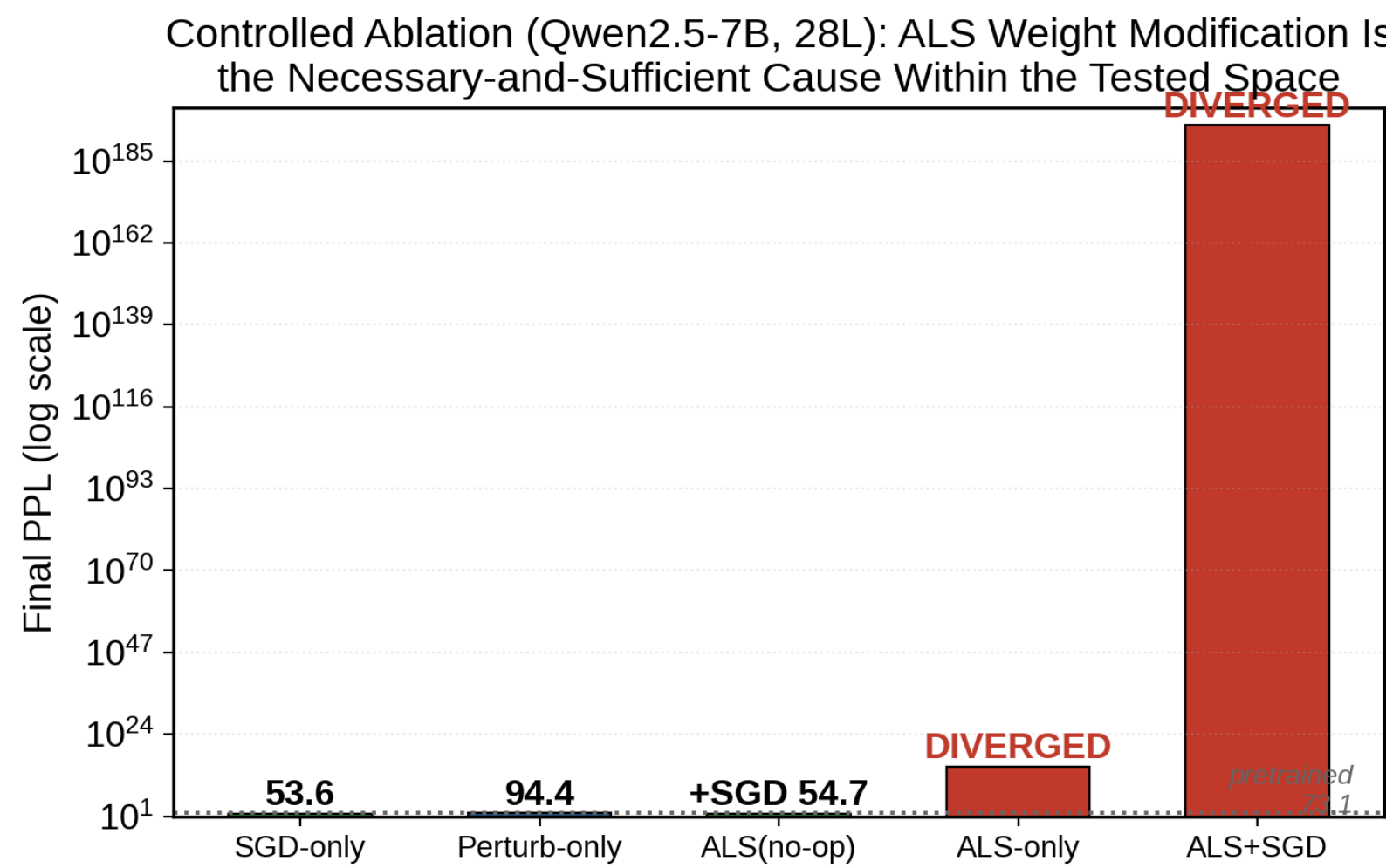
Finding 4 — 2×2 matrix: AdamW below baseline, ASP diverges

Qwen2.5-7B, 800 steps, N=3, all vs pretrained 73.1 PPL: AdamW full-rank 1.25, AdamW+LoRA 10.41, ASP+LoRA 135.4, ASP full-rank diverges.



Finding 5 — Ablation: ALS weight modification is the cause

ALS-only diverges in one solve ($73 \rightarrow 2.0 \times 10^9$ PPL); every ALS-free condition converges. Soft-target ALS (A-KD): 52.6 vs 52.4 — within noise, no evidence of benefit.



References

- [1] Hu, E. et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
 - [2] Loshchilov, I., Hutter, F. Decoupled Weight Decay Regularization. ICLR 2019.
 - [3] Koren, Y. et al. Matrix Factorization Techniques for Recommender Systems. IEEE Computer 42(8), 2009.
 - [4] Merity, S. et al. Pointer Sentinel Mixture Models. ICLR 2017.
 - [5] Hinton, G. et al. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.
- All experiments reproducible: runs/ + experiments/ in the project repo. Evaluation-harness audit and per-step data archived.